



Figure 8: Temperature evaluation in Celsius at the nodes

## Installation of Foren6 with Contiki

To configure the dependencies, type:

```
$ sudo apt-get install -y qt4-qmake libqt4-dev make
libxpat1-dev cmake libpcap0.8-dev libc6-dev tshark gcc g++
```

To get the source code with GIT, type:

```
$ git clone https://github.com/cetic/foren6.git
```

Use the following commands to install Foren6:

```
$ cd foren6
$ make
$ sudo make install
```

## Configuration of Sniffer to activate real-time capturing

To access and work on USB serial devices, the Linux Group 'dialout' is used. Initially, Foren6 is launched as the root user. If any other user account is used, then the user should be assigned the Linux Group 'dialout' to use USB.

```
$ sudo adduser <username> dialout
Sniffer Programming
```

### 6LoWPAN Troubleshooting with Foren6

Foren6 is an effort to provide a non-intrusive 6LoWPAN network analysis tool. It leverages passive sniffers to reconstruct a visual and textual representation of network information to support real-time Internet of Things applications where other means of debugging (e.g., in-network-based monitoring) are too costly or infeasible.

**Visualize your 6LoWPAN network.** Foren6 uses sniffers to capture 6LoWPAN traffic and renders the network state in a graphical user interface.

**Detect routing problems.** The Routing Protocol for 6LoWPAN (RPL) is an emerging IETF standard. Export statistics of RPL-related information and identify abnormal behaviors.

**On-site diagnosis.** Foren6 captures live packets from 6LoWPAN networks in a non-intrusive manner. Multiple sniffers can be combined for extended coverage.

**Customize to your infrastructure.** The network viewer uses flexible positions or user-defined locations to visualize 6LoWPAN networks in a 3D plot. It is ideal for debugging around a WSN.

**Android support.** An Android port is under development, allowing to visualize 6LoWPAN networks on a tablet. It is ideal for debugging around a WSN.

This project is maintained by CETIC. Find out more at [www.cetic.org.br](http://www.cetic.org.br).

Foren6 is Open Source. Start using Foren6 today at no cost. Follow the open-source development updates by subscribing to the foren6-dev mailing list.

Figure 9: Foren6 as a 6LoWPAN analyzer

```
$ git clone https://github.com/cetic/contiki
$ cd contiki
$ git checkout sniffer
$ make TARGET=sky savetarget sniffer.upload
```

## Scope of IoT based research simulations using Cooja

The Internet of Things is an emerging domain of research at various academic as well as corporate establishments. Because of the increasing number of devices currently connected under the IoT, there are vast areas for research in this domain.

The following are some of the approaches with which novel and effectual algorithms can be devised and implemented using Cooja:

- Interoperability and cross-protocol compatibility
- Development of energy-aware IoT scenarios
- Power aware scheduling and routing
- Prediction and avoidance of energy consumption attacks
- Lifetime analytics for robust IoT environments
- Reproducible and multi-interface implementations **END**

By: Dr Gaurav Kumar

The author is the MD of Magma Research and Consultancy Pvt Ltd, Ambala. He is associated with various academic and research institutes, where he delivers lectures and conducts technical workshops on the latest technologies and tools. You can contact him at [kumargaurav.in@gmail.com](mailto:kumargaurav.in@gmail.com). Website: [www.gauravkumarindia.com](http://www.gauravkumarindia.com).

THE COMPLETE MAGAZINE ON OPEN SOURCE

**OpenSource**  
ForYou

Your favourite Magazine on Open Source is now on the Web, too.

**OpenSourceForU.com**

Follow us on Twitter @LinuxForYou